

TRITON

v1.0

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Chapter 1

Bug List

File [main.cpp](#)

No known bugs.

Chapter 2

Hierarchical Index

2.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

ConfigUtils::arguments< T >	9
SuperTimer::ci_less	10
ExtBC::extBC< T >	11
Hydrograph::hydrograph< T >	12
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Chapter 3

Class Index

3.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

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SuperTimer::ci_less	10
DemFile::dem_file< T >	10
ExtBC::extBC< T >	11
Hydrograph::hydrograph< T >	12
Matrix::matrix< T >	12
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Chapter 4

File Index

4.1 File List

Here is a list of all documented files with brief descriptions:

config_utils.h	..	??
constants.h	..	??
dem_utils.h	..	??
extbc.h	..	??
inflow.h	..	??
kernels.h	..	??
main.cpp		
Main file containing the driver	..	17
matrix.h	..	??
mpi_utils.h	..	??
output.h	..	??
string_utils.h	..	??
supertimer.h	..	??
triton.h	..	??

Chapter 5

Class Documentation

5.1 ConfigUtils::arguments< T > Struct Template Reference

Public Attributes

- bool **time_increment_fixed**
- bool **time_series_flag**
- bool **gpu_direct_flag**
- int **checkpoint_id**
- int **num_sources**
- int **num_runoffs**
- int **num_extbc**
- int **it_count**
- T **time_step**
- T **sim_start_time**
- T **sim_duration**
- T **print_interval**
- T **courant**
- T **const_mann**
- T **hextra**
- std::string **outfile_pattern**
- std::string **hydrograph_filename**
- std::string **runoff_filename**
- std::string **print_option**
- std::string **input_format**
- std::string **output_format**
- std::string **output_option**
- std::string **dem_filename**
- std::string **src_loc_file**
- std::string **runoff_map**
- std::string **observation_loc_file**
- std::string **extbc_file**
- std::string **extbc_dir**
- std::string **h_infile**
- std::string **qx_infile**
- std::string **qy_infile**
- std::string **n_infile**
- std::vector< T > **src_x_loc**

- `std::vector< T > src_y_loc`
- `std::vector< T > observation_x_loc`
- `std::vector< T > observation_y_loc`
- `std::vector< T > extbc_x1_loc`
- `std::vector< T > extbc_y1_loc`
- `std::vector< T > extbc_x2_loc`
- `std::vector< T > extbc_y2_loc`
- `std::vector< int > extbc_bctype`
- `std::vector< std::string > extbc_fname`

The documentation for this struct was generated from the following file:

- `config_utils.h`

5.2 SuperTimer::ci_less Struct Reference

Classes

- struct [nocase_compare](#)

Public Member Functions

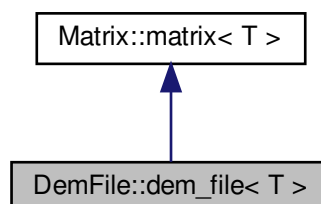
- `bool operator() (const std::string &s1, const std::string &s2) const`

The documentation for this struct was generated from the following file:

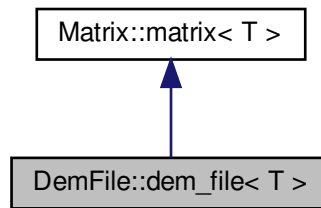
- `supertimer.h`

5.3 DemFile::dem_file< T > Class Template Reference

Inheritance diagram for `DemFile::dem_file< T >`:



Collaboration diagram for DemFile::dem_file< T >:



Public Member Functions

- **dem_file** (int rows, int cols)
- **dem_file** ([Matrix::matrix](#)< T > const &m)
- int **get_nrows** () const
- int **get_ncols** () const
- T **get_xll_corner** () const
- T **get_yll_corner** () const
- T **get_cell_size** () const
- int **get_no_data_value** () const
- void **set_nrows** (int row)
- void **set_ncols** (int col)
- void **set_xll_corner** (T xll)
- void **set_yll_corner** (T yll)
- void **set_cell_size** (T cell_size)
- void **set_no_data_value** (int no_data_value)
- void **load_header_from_dem_file_ascii** (std::string filename)
- void **load_header_from_dem_file_binary** (std::string filename)

The documentation for this class was generated from the following file:

- dem_utils.h

5.4 ExtBC::extBC< T > Class Template Reference

Public Member Functions

- **extBC** (std::string filename, int bctype)
- void **load_from_file** (std::string filename, int bctype)
- int **check_extreme_extbcb** (std::vector< int > e_cols, std::vector< int > e_rows, int ncols, int nrows)
- void **create_involved_cells** (std::vector< int > e_cols, std::vector< int > e_rows, int ncols, int nrows, int bctype)
- std::vector< std::vector< T > > **get_rows** ()
- T **get_var1_at** (int index)
- T **get_var2_at** (int index)
- int **get_num_rows** ()
- void **set_num_rows** (int rows)
- void **convert_to_secs** ()

Public Attributes

- int **ncells**
- int **location**
- int **ncells_local**
- std::vector< int > **extreme_rows**
- std::vector< int > **extreme_cols**
- std::vector< int > **i_cols**
- std::vector< int > **i_rows**

The documentation for this class was generated from the following file:

- extbc.h

5.5 Hydrograph::hydrograph< T > Class Template Reference

Public Member Functions

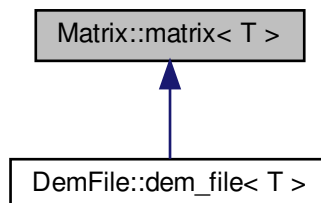
- **hydrograph** (std::string filename)
- void **load_from_file** (std::string filename)
- std::vector< std::vector< T > > **get_rows** ()
- T **get_flow_at** (int index, int source_num)
- T **get_time_at** (int index)
- int **get_num_inflow_rows** ()
- int **get_num_inflows** ()
- void **convert_time_hr_to_secs** ()
- void **convert_rate_hr_to_secs** ()
- void **convert_rate_mm_to_m** ()
- void **set_num_flow_rows** (int rows)
- void **set_num_sources** (int sources)

The documentation for this class was generated from the following file:

- inflow.h

5.6 Matrix::matrix< T > Class Template Reference

Inheritance diagram for Matrix::matrix< T >:



Public Member Functions

- **matrix** (int rows, int cols)
- **matrix** (int rows, int cols, T **arr)
- **matrix** ([matrix](#)< T > const &m)
- T & **operator()** (int row, int col)
- T **operator()** (int row, int col) const
- [matrix](#) & **operator=** ([matrix](#)< T > m)
- [matrix](#) & **operator*=** (T value)
- [matrix](#) & **operator*** (T value)
- [matrix](#) & **operator+** (T value)
- [matrix](#) & **operator+** ([matrix](#) const &m)
- [matrix](#) & **operator+=** ([matrix](#) const &m)
- [matrix](#) & **operator+=** (T value)
- [matrix](#)< T > & **operator*** ([matrix](#) const &m)
- T * **get_data** () const
- T * **begin** ()
- T * **get_address_at** (int row, int col)
- int **get_num_rows** () const
- int **get_num_cols** () const
- int **get_ghost_nrows** () const
- int **get_ghost_ncols** () const
- void **set_size** (int rows, int cols)
- void **resize** (int rows, int cols)
- void **set_value** (int row, int col, T value)
- void **set_value** (std::pair< int, int > cell, T value)
- void **set_value** (int index, T value)
- T **get_value** (int row, int col)
- T **get_value** (std::pair< int, int >)
- T **get_value** (int index)
- void **add_ghost_cells** (int grows, int gcols, T value)
- void **remove_ghost_cells** ()
- void **copy_value_into_ghost_cells** ()
- void **copy_elevation_into_ghost_cells** (std::vector< int > irows, std::vector< int > icols, int ncells, int location)
- void **set_infinite_walls** ()
- bool **is_inbounds** (int row, int col)
- void **zero_fill** ()
- void **zero_fill_int** ()
- void **pow** (T e)
- void **square** ()
- void **load_from_ascii_file** (std::string &filepath)
- void **load_from_ascii_file** (int rows, int cols, std::string &filepath)
- void **load_from_ascii_file** (int rows, int cols, std::string &filepath, int header_size)
- void **load_from_binary_file** (int rows, int cols, std::string &filepath)
- void **load_from_binary_file** (int rows, int cols, std::string &filepath, int header_size)
- std::pair< int, int > **get_dims_2d** (std::string &filepath)

The documentation for this class was generated from the following file:

- [matrix.h](#)

5.7 SuperTimer::ci_less::nocase_compare Struct Reference

Public Member Functions

- **bool operator()** (const unsigned char &c1, const unsigned char &c2) const

The documentation for this struct was generated from the following file:

- supertimer.h

5.8 Output::output< T > Class Template Reference

Public Member Functions

- void **init** (int rows, int cols, int rank, int size, std::string project_dir, std::string outfile_pattern, int time_series↵_flag, std::string cfg_content, std::string output_option)
- void **init_time_series** (int observation_loc_size, Constants::sources_list_t observation_cells)
- void **write_output** (Matrix::matrix< T > &h_arr, Matrix::matrix< T > &qx_arr, Matrix::matrix< T > &qy_arr, std::string output_format, std::string print_option, int print_id, int it_count, T simtime, T global_dt)
- void **write_output_ascii_sequential** (Matrix::matrix< T > &arr, std::string what_mat, int print_id, T simtime)
- void **write_output_ascii_parallel** (Matrix::matrix< T > &arr, std::string what_mat, int print_id, T simtime)
- void **write_output_binary_sequential** (Matrix::matrix< T > &arr, std::string what_mat, int print_id, T simtime)
- void **write_output_binary_parallel** (Matrix::matrix< T > &arr, std::string what_mat, int print_id, T simtime)
- std::string **get_mat_path** (std::string what, std::string subdir, int print_id, std::string extension)
- void **output_time_series** (std::string what_mat, int print_id, T simtime)
- void **output_cfg** (T simtime, int print_id, T global_dt, int it_count)
- void **write_times** (SuperTimer::super_timer st)
- double **average** (double a[], int n)

The documentation for this class was generated from the following file:

- output.h

5.9 MpiUtils::partition_data_t Struct Reference

Public Member Functions

- **partition_data_t** (int s, int r, int c)

Public Attributes

- int **size**
- int **rows**
- int **cols**
- int **sub_rows**
- int **sub_cols**
- int **cart_dims** [2]
- std::vector< Constants::dims_t > **part_dims**

The documentation for this struct was generated from the following file:

- mpi_utils.h

5.10 SuperTimer::super_timer Class Reference

Public Member Functions

- void **start** (std::string category)
- void **stop** (std::string category)
- void **reset** ()
- double **get_total_time** ()
- double **get_custom_time** (std::string category)
- std::string **get_current_date** ()
- std::string **get_hostname** ()
- int **add_new_timer** (std::string category)

The documentation for this class was generated from the following file:

- supertimer.h

5.11 Triton::triton< T > Class Template Reference

Public Member Functions

- **triton** (int argc, char *argv[])
- void **initialize** (int rank_, int size_)
- void **simulate** ()

The documentation for this class was generated from the following file:

- triton.h

Chapter 6

File Documentation

6.1 main.cpp File Reference

Main file containing the driver.

```
#include <iostream>
#include <string>
#include <float.h>
#include <utility>
#include <map>
#include <fstream>
#include <vector>
#include <sys/time.h>
#include <sys/stat.h>
#include <unistd.h>
#include <algorithm>
#include <sstream>
#include <iomanip>
#include <dirent.h>
#include <cmath>
#include <cstring>
#include <omp.h>
#include "mpi.h"
#include "constants.h"
#include "supertimer.h"
#include "string_utils.h"
#include "mpi_utils.h"
#include "config_utils.h"
#include "inflow.h"
#include "extbc.h"
#include "matrix.h"
#include "dem_utils.h"
#include "output.h"
#include "triton.h"
#include "kernels.h"
```


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